

Python Basics: Assignment Submission

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Subject: Python Scripting & File Handling

1. Grade Checker

Objective: Take a score as input and print the grade based on specific ranges using if-elif-else statements.

Code:

```
Python
# Grade Checker Program

score = int(input("Enter your score: "))

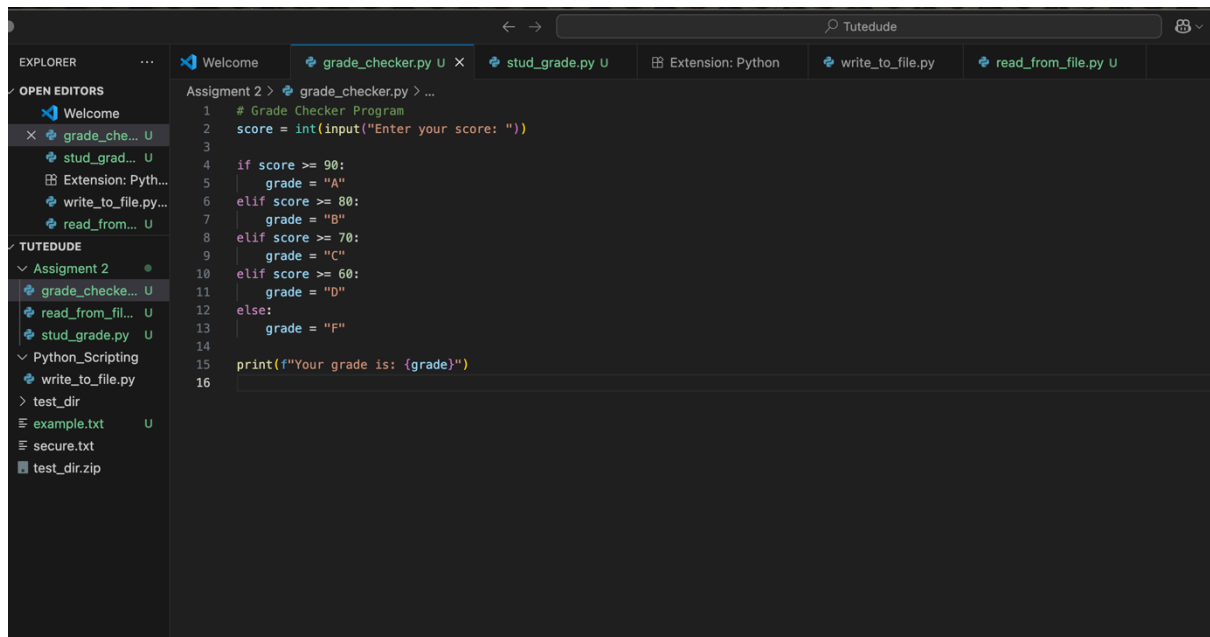
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
elif score >= 60:
    print("Grade: D")
else:
    print("Grade: F")
```

Explanation:

- The program takes user input using input().
- It converts the input string to an integer using int().
- It uses if-elif-else conditional logic to check the score against the defined ranges and prints the corresponding grade.

Sample Output:

```
Plaintext
Enter your score: 85
Grade: B
```



2. Student Grades Dictionary

Objective: Create a menu-driven program using a dictionary to store student names and grades, allowing the user to add, update, and print grades.

Code:

Python

Student Grades Management

```
students = {}
```

```
while True:
```

```
    print("\n1. Add Student")
    print("2. Update Student Grade")
    print("3. Display All Students")
    print("4. Exit")
```

```
    choice = input("Enter your choice: ")
```

```
    if choice == '1':
```

```
        name = input("Enter student name: ")
        grade = input("Enter grade: ")
        students[name] = grade
        print(f"Added {name} with grade {grade}")
```

```
    elif choice == '2':
```

```
        name = input("Enter student name to update: ")
        if name in students:
            grade = input("Enter new grade: ")
            students[name] = grade
            print(f"Updated {name}'s grade to {grade}")
        else:
```

```
        print("Student not found.")

elif choice == '3':
    print("\nStudent Grades:")
    for name, grade in students.items():
        print(f"{name}: {grade}")

elif choice == '4':
    break

else:
    print("Invalid choice!")
```

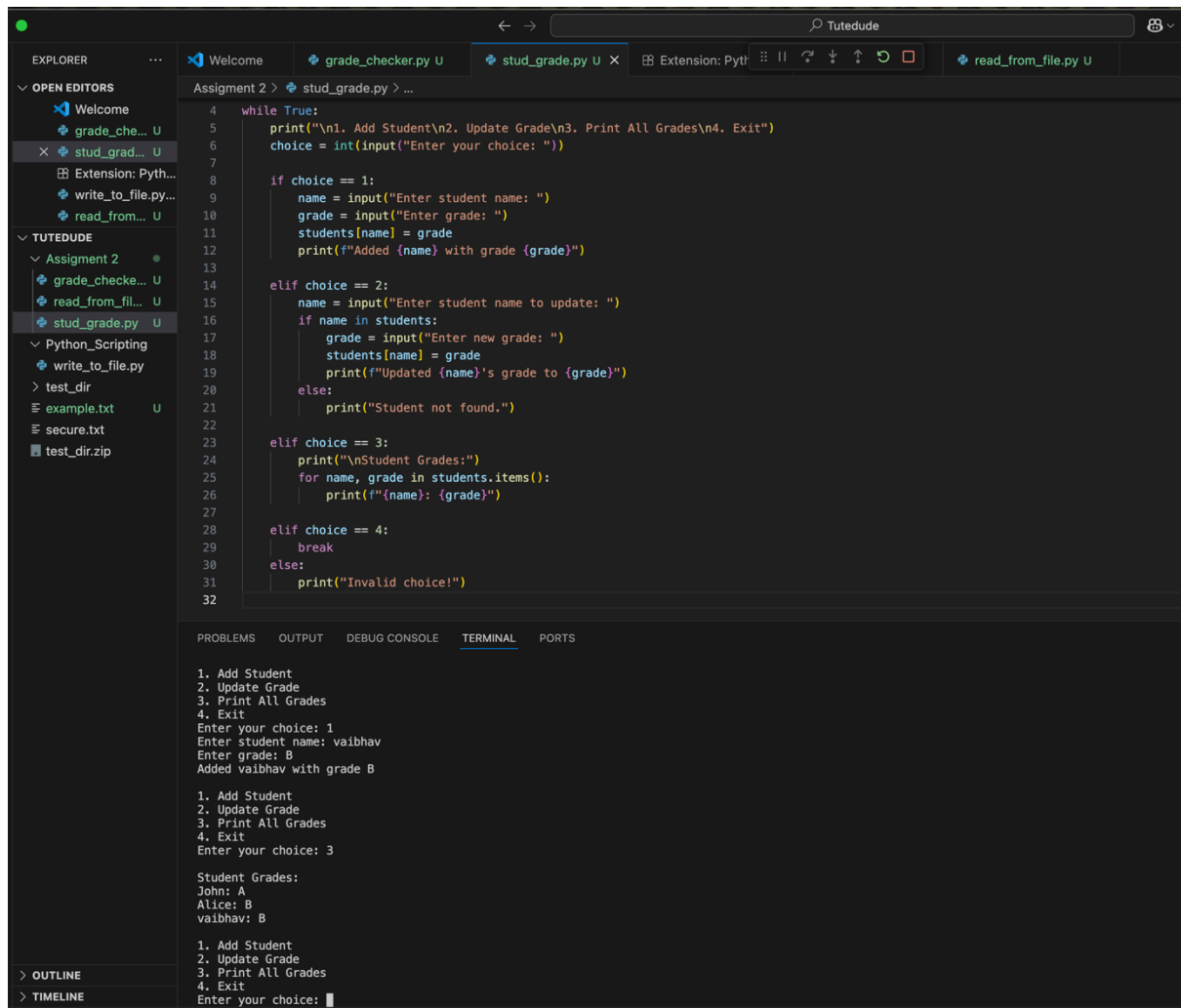
Explanation:

- Initializes an empty dictionary called `students` to store the data.
- Uses a `while True` loop to create a continuous menu-driven interface.
- **Add:** Assigns a new key-value pair via `students[name] = grade`.
- **Update:** Checks if the student exists using `if name in students` before updating.
- **Display:** Iterates through the dictionary using a `for` loop and the `.items()` method.

Sample Output:

Plaintext

```
1. Add Student
2. Update Grade
3. Print All Grades
4. Exit
Enter your choice: 1
Enter student name: vaibhav
Enter grade: B
Added vaibhav with grade B
```



3. Write to a File

Objective: Write a program to create a text file and write content to it.

Code:

Python

Write to File Program

```
file = open("example.txt", "w") # open in write mode
file.write("Hello, this is a test file.\nThis is the second line.")
file.close()
```

```
print("Data written to example.txt")
```

Explanation:

- `open("example.txt", "w")`: Opens (or creates) the file in write mode (w).

- `file.write()`: Inserts the specified string content into the text file. The `\n` character is used to create a new line.
- `file.close()`: Closes the file, ensuring the changes are saved properly and system resources are freed.

Sample Output:

Plaintext

Data written to example.txt

The screenshot shows a VS Code editor with a Python script named `write_to_file.py` open. The script contains the following code:

```
1 # Write to File Program
2 file = open("example.txt", "w") # open in write mode
3 file.write("Hello, this is a test file.\nThis is the second line.")
4 file.close()
5
6 print("Data written to example.txt")
7
```

The Explorer sidebar on the left shows the file structure, including a directory named `test_dir` which contains `example.txt`, `secure.txt`, and `test_dir.zip`.

The Terminal at the bottom shows the output of the script:

```
/usr/bin/env /usr/local/bin/python3 /Users/vaibhavkarkute/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-x64/bundled/libs/debugpy/adapter/launcher 49415 -- /Users/vaibhavkarkute/Learning/Tutedude/Assignment\ 2/read_from_file.py
vaibhavkarkute@MacBookPro Tutedude % /usr/bin/env /usr/local/bin/python3 /Users/vaibhavkarkute/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-x64/bundled/libs/debugpy/adapter/launcher 49415 -- /Users/vaibhavkarkute/Learning/Tutedude/Assignment\ 2/read_from_file.py
File content:
Hello, this is a test file.
This is the second line.
vaibhavkarkute@MacBookPro Tutedude %
```

4. Read from a File

Objective: Read the contents of the previously created text file and display them in the console.

Code:

Python

Read from File Program

```
file = open("example.txt", "r")
content = file.read()
print("File content:")
```

```
print(content)
file.close()
```

Explanation:

- `open("example.txt", "r")`: Opens the file in read mode (r).
- `file.read()`: Reads the entire content of the file and stores it in the `content` variable.
- The script then prints the stored string to the terminal.

Sample Output:

Plaintext
File content:
Hello, this is a test file.
This is the second line.